SWIR/NIR Spectral Micro-Beads for Autonomous Search and Rescue Detection

A Speculative Framework and Feasibility Assessment

Collaboration Whitepaper

Not a Funding Request

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Note: The author is an independent systems engineer without institutional affiliation or prior publication record in optical physics. This work represents a concept-stage proposal seeking domain expertise to evaluate feasibility.

1. Introduction

Search and rescue operations depend on the ability to distinguish a casualty from the surrounding environment—rapidly, reliably, and across a range of conditions. The detection technologies currently in service each carry well-characterised failure modes that constrain their operational envelopes. Thermal imaging in the long-wave infrared (LWIR) band loses contrast as a casualty's surface temperature converges with ambient, and it cannot penetrate solid occlusions such as debris or dense canopy. Visible-spectrum cameras, even when paired with trained detection models, degrade in poor visibility and cluttered scenes. Broadband retroreflective materials—the basis for current SOLAS/IMO-compliant personal protective equipment—require active illumination and return all illuminated surfaces without spectral discrimination, producing background clutter alongside the target signal.

These limitations share a common thread: none of the existing approaches encodes a distinctive spectral signature onto the casualty that could be isolated from environmental noise by a suitably equipped detection platform. All current SOLAS/IMO standards specify retroreflection performance exclusively in the visible band, which means the regulatory framework does not yet account for drone-based detection systems operating in the short-wave infrared (SWIR) or near-infrared (NIR) regions.

This paper describes a speculative concept intended to address that gap: rigid glass or polymer micro-beads, coated with a multilayer thin-film structure, tuned to reflect in a narrow SWIR/NIR band between 900 and 1700 nm. The multilayer design draws on the physical principle underlying labradorite iridescence—alternating layers of differing refractive index producing constructive interference at specific wavelengths—adapted conceptually to engineered substrates. The beads would create spectral signatures detectable by drone-mounted InGaAs cameras equipped with bandpass filters, separating the target signal from broadband environmental scatter.

A critical caveat must be stated at the outset. The proposed beads are spherical, and a sphere does not produce coherent retroreflection; it scatters reflected light over a wide angular range. A preliminary order-of-magnitude estimate, presented in Section 3, suggests that the geometric scattering loss from spherical geometry is likely to exceed the spectral discrimination gain from narrow-band filtering. The spherical form factor is the simplest starting geometry for analysis, not a fixed design requirement. Whether this concept survives first-principles modelling is an open question—one that this paper is designed to pose, not to answer. ***[HYPOTHESIS]***

2. Background

2.1 Existing Detection Technologies and Their Failure Modes

Thermal imaging in the LWIR band exploits the temperature differential between a living casualty and the surrounding environment. This works well when the differential is large—a warm body against cold ground—but the advantage diminishes as the casualty's surface temperature approaches ambient. Prolonged exposure, wet clothing, and hypothermia all drive this convergence. Solid occlusions present an absolute

barrier: LWIR radiation does not pass through debris, building materials, or dense vegetation canopy.

Visible-spectrum detection, whether human-operated or model-assisted, performs well in clear conditions with unobstructed sightlines. Dense vegetation, low-contrast scenes, poor lighting, and adverse weather conditions all degrade performance substantially. Model-assisted systems are only as good as their training data, and false-positive rates in cluttered environments remain a persistent challenge for operational deployment.

Broadband retroreflective materials—most commonly encountered as SOLAS-grade retroreflective tape on maritime personal protective equipment—work by returning incident light back towards its source. The operative word is “broadband.” These materials return all illuminated surfaces, target and background alike, without any spectral discrimination. The signal-to-noise ratio depends entirely on spatial resolution and scene geometry, not on any intrinsic property of the marker itself.

2.2 The Regulatory Gap

All current SOLAS and IMO standards define retroreflection performance in the visible band. No existing specification addresses SWIR or NIR detection capability. Introducing SWIR detectability requirements into the IMO standards framework would require proposal submission, working group review, flag state consultation, and plenary adoption—a process estimated to take five to eight years. This regulatory timeline is relevant because any maritime application of the proposed concept depends on standards evolution, regardless of its technical merit.

2.3 The Proposed Concept

The concept under evaluation uses micro-beads in a notional diameter range of 50–500 µm (optimal size remains an open question) coated with a low-refractive-index-difference multilayer thin-film structure. The coating is designed to produce constructive interference—and therefore selective reflection—at a specific narrow SWIR wavelength band. When illuminated by a SWIR source mounted on a detection platform, the spectral reflection signature would, in principle, separate the target from broadband environmental scatter.

The spectral reflection band lies above approximately 700 nm and would not be visible to the naked eye, though the beads themselves remain physically visible objects. Any sideband leakage from the multilayer structure below 700 nm could produce visible iridescence—an artefact, not a feature.

Wavelength multiplexing—using different reflection bands to encode information such as unit identification or triage priority—is a theoretical extension. It would require fabricating multiple bead populations with spectrally distinct, narrow, non-overlapping reflection bands and sufficient spectral separation for reliable field discrimination. The fabrication precision required for multiplexing significantly exceeds that for a single-band system. This should be considered a second-generation capability, contingent on demonstrating single-band feasibility first.

3. The Conceptual Framework

3.1 Novelty Assessment

A prior-art search was conducted across academic literature, patents (USPTO, EPO, WIPO), military procurement databases, and commercial product catalogues during January and February 2026. This search was limited to unclassified sources and may therefore be incomplete. No prior art was found combining structural-colour multilayer micro-beads with SWIR/NIR wavelengths for search and rescue detection.

The absence of prior art should not be mistaken for evidence of value. It may equally indicate that the approach has been evaluated and found non-viable by others, or that similar concepts exist within classified programmes. The closest conceptual analogue in the open literature is the photonic crystal encoded beads described by Gu et al. (2013) for laboratory bio-assays, which operate in the visible band for in-vitro applications and are not designed for environmental durability or remote detection. Military identification-friend-or-foe technology in unclassified literature is RF-transponder-based and does not use passive spectral signatures.

3.2 Preliminary Radiometric Estimate

The following order-of-magnitude estimate is provided to frame the feasibility question, not to claim a specific performance level. ***[HYPOTHESIS]***

On the spectral discrimination side: if the bead reflects in a 40 nm band within the 800 nm SWIR window spanning 900–1700 nm, the spectral discrimination ratio is approximately twenty-fold. This 40 nm bandwidth is assumed, not demonstrated. Multilayer thin-film structures on flat substrates routinely achieve bandwidths of 10–50 nm at full-width half-maximum, but conformal deposition on curved micro-bead surfaces may produce broader, less well-defined reflection peaks due to varying layer thickness across the bead curvature. The actual bandwidth—and therefore the actual spectral gain—depends on fabrication quality that has not been characterised.

On the geometric side: a smooth sphere illuminated by a collimated beam distributes reflected energy over approximately 2π steradians. For a drone-mounted detector at 100 m altitude with a 50 mm aperture lens, the solid angle subtended by the detector is on the order of 10^{-7} steradians. The fraction of reflected energy reaching the detector is therefore roughly 10^{-7} to 10^{-8} of the total reflected energy. An engineered retroreflector—corner cube or cat's-eye—directs reflected energy into a narrow cone, typically 0.2° to 1° half-angle, concentrating returned energy by a factor of ten thousand to one million relative to a diffuse sphere.

The geometric scattering loss exceeds the spectral discrimination gain by several orders of magnitude. A spherical bead with a narrow-band multilayer coating is expected to return substantially less energy to a co-located detector than a broadband retroreflector, even after accounting for spectral filtering on the detector side. This estimate omits several factors—bead array effects, near-field scattering contributions from bead clusters, and specular highlights from Fresnel reflection at normal incidence on the sphere apex—but the magnitude of the geometric disadvantage strongly suggests that

simple spherical beads are unlikely to outperform existing retroreflectors in co-located illumination and detection geometries. ***[HYPOTHESIS]***

3.3 Rationale for Continued Investigation

Given the adverse preliminary estimate, the question becomes why continued investigation is warranted rather than immediate abandonment. Four arguments support further modelling before concluding.

Non-spherical geometries may recover the geometric loss. Flattened platelet beads, hemispheres with reflective backing, or microspheres with embedded retroreflective cavities could combine spectral selectivity with directional return. The spectral coating concept is separable from the spherical form factor, and the preliminary estimate applies specifically to spheres, not to the coating concept itself.

Bistatic detection geometries change the link budget. If the detection platform and the illumination source are not co-located, the retroreflective advantage of corner cubes diminishes and the comparison with spherical scattering becomes less unfavourable. Whether operationally practical bistatic configurations exist for drone-based search and rescue is itself an open question.

Spectral coding provides information content that retroreflection does not. Even if radiometric performance is inferior, a spectrally encoded marker provides identification capability—distinguishing who or what is being detected—that a broadband retroreflector cannot. Whether this information value justifies the signal penalty is an operational question, not a physics question.

The concept may find applications where retroreflectors are impractical. Beads can be embedded in textiles, distributed over surfaces, or mixed into coatings—form factors that are incompatible with rigid retroreflector elements. If the operational requirement is “some detection capability in a flexible form factor” rather than “maximum detection range,” the trade-off may be acceptable.

These are reasons to model the concept properly, not reasons to assume it works. If detailed modelling confirms the preliminary estimate with no viable geometry or detection architecture, the concept should be abandoned.

3.4 Target Applications

All applications described below are conditional on two prerequisites: radiometric modelling demonstrating net positive detection performance in at least one viable geometry, and fabrication feasibility at the required spectral precision. Given the adverse preliminary estimate, these applications should be read as motivational context for the modelling effort, not as near-term deployment targets. ***[HYPOTHESIS]***

In maritime search and rescue, SWIR-detectable markers on SAR equipment would enable drone-based casualty location. This application would require SOLAS/IMO standards evolution over a five-to-eight-year process, and is relevant only if detection performance exceeds existing SOLAS retroreflectors.

For military combat search and rescue, casualty marking with spectral signatures outside the visible band would be detectable by friendly SWIR-equipped platforms while remaining invisible to the unaided eye. The detection platform requires an active SWIR illuminator; the beads are passive markers.

Wilderness and mountain rescue presents a scenario where spectrally marked clothing or equipment could be detectable through forest canopy. SWIR canopy penetration is better than visible-band penetration, though still subject to significant attenuation and scattering.

Disaster response offers a potential use case for spectral multiplexing to encode triage priority levels, though this depends entirely on multiplexing fabrication feasibility.

4. Discussion and Limitations

4.1 Spherical Geometry Efficiency Loss

This is the concept's most significant vulnerability. A sphere distributes reflected energy over approximately 2π steradians, and the preliminary estimate indicates the geometric scattering loss exceeds the spectral discrimination gain by four to six orders of magnitude. For co-located illumination and detection geometries, signal-to-noise inversion—where the spectrally selective bead system performs worse than a broadband retroreflector—is the expected outcome for simple spherical beads. ***[HYPOTHESIS]***

The spherical form factor was initially proposed because it is the natural geometry for micro-beads used in coatings and textiles. The physics, however, strongly suggests that alternative geometries—platelet beads, hemispheres with reflective backing, microspheres with retroreflective cavities—should be the primary focus of any modelling effort. The viability of the spectral approach should be evaluated across multiple form factors before drawing conclusions about feasibility.

4.2 Fabrication Constraints

Conformal multilayer deposition on micro-bead substrates—whether spherical or non-spherical—with individual layer thickness control at 250–375 nm sits at or beyond the current state of the art for polymer co-extrusion. The target is polymer co-extrusion using materials such as PEN and PMMA, with a refractive index difference of approximately 0.17–0.26. Even if an optical model identifies a geometrically viable configuration, fabrication at the required precision and cost may not be achievable. This is the second critical feasibility gate after the radiometric model.

4.3 Environmental Durability

SOLAS equipment must withstand extended saltwater immersion. Polymer multilayer structures face delamination, osmotic ingress, and degradation in marine environments. Durability testing is required for any maritime application, though it sits downstream of the radiometric and fabrication feasibility gates—there is little value in testing durability for a concept that fails on physics or fabricability.

4.4 Atmospheric and Scene-Level Spectral Interference

SWIR atmospheric transmission contains multiple absorption features, with water vapour bands at approximately 940, 1130, and 1380 nm constraining the usable wavelength space. Natural scene elements—minerals, wet vegetation, water surfaces—have their own SWIR spectral features that could produce false-positive detections. The bead reflection band must be co-designed with atmospheric transmission windows and the spectral clutter characteristics of the target operational environment. Neither constraint has been quantitatively addressed in this work.

4.5 Standards and Regulatory Timeline

Introducing SWIR detectability requirements into IMO standards requires an estimated five to eight years of standards development activity. The concept cannot be deployed in SOLAS personal protective equipment without this process completing. For non-maritime applications, the regulatory pathway may be shorter but remains undefined.

4.6 Collaboration Recruitment

This project requires expertise spanning multilayer thin-film deposition on curved substrates, SWIR spectral imaging, drone integration, search and rescue operations, and IMO standards navigation. The author possesses systems engineering experience but not domain expertise in any of these specialisms. The project has no institutional backing, no funding, and no preliminary experimental data. The probability of recruiting the required multidisciplinary team through outreach from this paper is acknowledged to be low. The paper is circulated on the basis that the cost of circulation is negligible, not on the expectation of high response rates.

5. Conclusion

5.1 Open Questions Requiring Collaboration

The questions below are ordered by criticality. The first two are gate questions—negative answers terminate the concept.

The fundamental feasibility question is radiometric: what is the detection performance of a multilayer-coated bead—in both spherical and alternative geometries—compared with an engineered retroreflector across operationally relevant scenarios? The preliminary estimate suggests spherical geometry is non-viable; modelling should explore non-spherical alternatives and bistatic detection architectures. This requires expertise in optical engineering and radiometric modelling.

The second gate question concerns fabrication: can multilayer thin-film structures be conformally deposited on micro-bead substrates in the 50–500 μm diameter range with sufficient layer thickness uniformity to produce narrow-band SWIR reflection? This requires expertise in thin-film deposition and polymer co-extrusion.

Beyond the gate questions, three further areas demand investigation: optimal SWIR wavelength band selection accounting for atmospheric transmission and scene-level

spectral clutter; detection hardware specification in a drone-deployable form factor, including sensor, filter, and illuminator selection; and environmental durability characterisation for marine and terrestrial exposure conditions.

5.2 Invitation to Collaborate

This paper is issued as an open invitation to research collaboration. The project is not seeking funding at this stage. The expertise sought spans optical engineering and radiometric modelling (the critical path), thin-film deposition and polymer co-extrusion on curved substrates, SWIR/NIR imaging and spectral detection systems, drone-based remote sensing and search and rescue operations, and maritime safety standards.

The single most valuable immediate contribution would be a radiometric model evaluating the spectral-gain-versus-geometric-loss trade-off across spherical and non-spherical bead geometries. That model would determine whether the concept merits any further investigation, or whether it should be set aside.

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6. References

Gu, H. et al. (2013) *Photonic Crystal Encoded Microcarriers for Biomolecule Detection.* Small, 9(13), 2266–2271.

IMO Resolution MSC.81(70) — Testing and evaluation of life-saving appliances (SOLAS standard).

Appendix A: Document Metadata

Field	Value
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Appendix B: Glossary

Abbreviation / Term	Definition
SWIR	Short-Wave Infrared; wavelength range approximately 900–1700 nm
NIR	Near-Infrared; wavelength range approximately 700–1000 nm
LWIR	Long-Wave Infrared; wavelength range approximately 8–14 µm, used for thermal imaging
InGaAs	Indium Gallium Arsenide; semiconductor material used in SWIR-sensitive photodetectors
SOLAS	Safety of Life at Sea; international maritime safety treaty administered by IMO
IMO	International Maritime Organisation; United Nations agency for maritime safety and environmental regulation
SAR	Search and Rescue
IFF	Identification Friend or Foe; systems used for military target identification
FWHM	Full-Width at Half-Maximum; standard measure of spectral bandwidth
SNR	Signal-to-Noise Ratio
Δn	Refractive index difference between alternating layers in a multilayer thin-film structure
PEN	Polyethylene Naphthalate; polymer used in multilayer optical films
PMMA	Polymethyl Methacrylate; polymer used as a low-index layer in multilayer optical films
Retroreflector	Optical device that reflects incident light back towards its source with minimal scattering
Bistatic	Detection geometry where the illumination source and detector are at different positions
Co-located	Detection geometry where the illumination source and detector share the same position
Conformal deposition	Thin-film coating process that follows the surface contours of the substrate
Spectral clutter	Background spectral features from natural or artificial scene elements that may mimic the target signature
Bandpass filter	Optical filter that transmits only a specified wavelength range, rejecting all others

Appendix C: Methodology and Transparency Statement

Author Context and Expertise

This work is grounded in 30 years of multi-sector experience in Systems Engineering. The analytical framework reflects professional history, focusing on practical application.

Digital Methodology and Accessibility

In the spirit of transparency, I utilised a suite of Generative AI tools—specifically Claude, Google Gemini, and ChatGPT—to assist in the production of this paper. These tools were employed as “force multipliers” for data synthesis and editorial accessibility. As a writer with dyslexia, I use these models as assistive technology to refine grammar and streamline sentence structure.

Verification and Integrity

Whilst AI assisted in processing data, the intellectual oversight is entirely human. I performed manual validation of all citations and accept full responsibility for the accuracy and originality of the final output.

Appendix D: Raw Author Notes

The original concept for spectrally selective micro-beads emerged from Project Labradorite, a research effort investigating the commercial potential of synthetic labradorescence—the iridescent optical phenomenon observed in labradorite feldspar. The feldspar's iridescence arises from alternating lamellae of differing composition (the Bøggild intergrowth structure) that produce thin-film interference effects. The question driving the SAR application was whether this natural optical principle could be adapted to engineered polymer or glass substrates at the micro-bead scale, targeting wavelengths outside the visible band.

Earlier drafts used the term "Bøggild-mimetic" to describe the multilayer coating concept. This terminology was retired because it is not standard in the thin-film optics literature and implied a more direct correspondence between natural labradorite microstructure and engineered polymer multilayers than is warranted. The current text uses descriptive language ("low-Δn multilayer thin-film structure") throughout.

The author's background is in systems engineering across multiple sectors. Domain expertise in optical physics, thin-film deposition, and SWIR imaging is explicitly absent from the project team. This paper exists precisely to attract that expertise.

Appendix E: Change Log

This appendix documents editorial and structural decisions made during the preparation of this paper.

Terminology changes:

The term “Bøggild-mimetic” was replaced throughout with descriptive language referencing “low-Δn multilayer thin-film structure.” The original term, whilst etymologically accurate, is not standard in the thin-film optics literature and could mislead readers about the specificity of the analogy between natural labradorite and the proposed engineered coatings.

Section labelling:

The opening section was labelled “Abstract” in earlier working drafts. For the final document, the content was redistributed across a full Introduction and structured body sections to provide the narrative flow expected in a collaboration whitepaper rather than a journal manuscript.

Radiometric estimate:

A preliminary order-of-magnitude radiometric estimate was incorporated into Section 3 to address the concern that the paper, without quantitative framing, relied on hand-waving. The estimate indicates the geometric scattering loss from spherical beads likely exceeds the spectral discrimination gain, motivating investigation of non-spherical geometries.

Application framing:

Target applications were reframed as motivational context for the modelling effort rather than near-term deployment targets, reflecting the concept’s pre-feasibility status.

Signal-to-noise characterisation:

The SNR inversion risk for spherical beads was upgraded from “possible” to “probable,” reflecting the quantitative estimate’s findings. Modelling recommendations were updated to focus on geometries and detection architectures where inversion does not occur.

Collaboration recruitment:

An explicit acknowledgement of the low probability of successful cold-outreach recruitment was added to the Discussion and Limitations section, framing the paper's circulation as a low-cost effort rather than a confident expectation of team assembly.